

Lab 3: Architectural Decision Justification

Component Modelling & Architectural Pattern Selection

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Problem Statement: #53: Freelance Content Creator Escrow Platform (Media, Events & Community)

Architecture Selection: We chose **Microservices Architecture** for the Freelance Content Creator Escrow Platform System.

Reason 1 - Divergent Performance Profiles Across Workflow Stages

The platform combines workloads with very different traffic profiles: the Community & Events Service experiences sharp, unpredictable spikes during live streams, contests, and networking events, while the Escrow & Payment Service carries a steadier, transaction-critical load. A microservices approach lets each service be scaled and provisioned independently on its own infrastructure, so a viral community event never starves payment processing of resources, and the payment path is never over-provisioned just to absorb community traffic bursts.

Reason 2 - Isolating the Approval/Payment Sign-off Gate

Freelancers depend on the platform to hold and release real money, so the Escrow & Payment Service must remain available and correct even if other parts of the system fail. Splitting the system into independently deployable services (Profile, Contracts, Escrow, Community, Notification) means a bug or outage in the Community & Events Service cannot take down escrow processing, and each team can release updates to their own service without redeploying the whole platform.

Security Advantage

Isolating the Escrow & Payment Service into its own microservice with a narrow, well-defined API shrinks the compliance boundary (e.g., PCI-DSS scope) to a single component instead of the entire application. Sensitive payment credentials and escrow ledgers are never exposed to the Community, Profile, or Notification services, so a vulnerability discovered in the public-facing community features still can't access sensitive financial data, keeping the impact of a breach contained.

Performance Benefit

Because the Escrow & Payment Service publishes a "payment released" event that the Notification Service consumes asynchronously (rather than the escrow flow blocking on sending emails/push alerts), the user-facing payment confirmation returns quickly while notifications are dispatched in the background. Combined with independent horizontal scaling of the Community & Events Service during peak live-event traffic, the architecture keeps critical payment latency low without over-provisioning the entire system.